Chapter 3, Problem 78P

Problem

Solve Prob. 3.20 using *PSpice* or *MultiSim*.

Step-by-step solution

Step 1 of 10

Refer to Figure 3.69 in the text book.

Identify the nodes and voltage in the circuit shown in Figure 1.

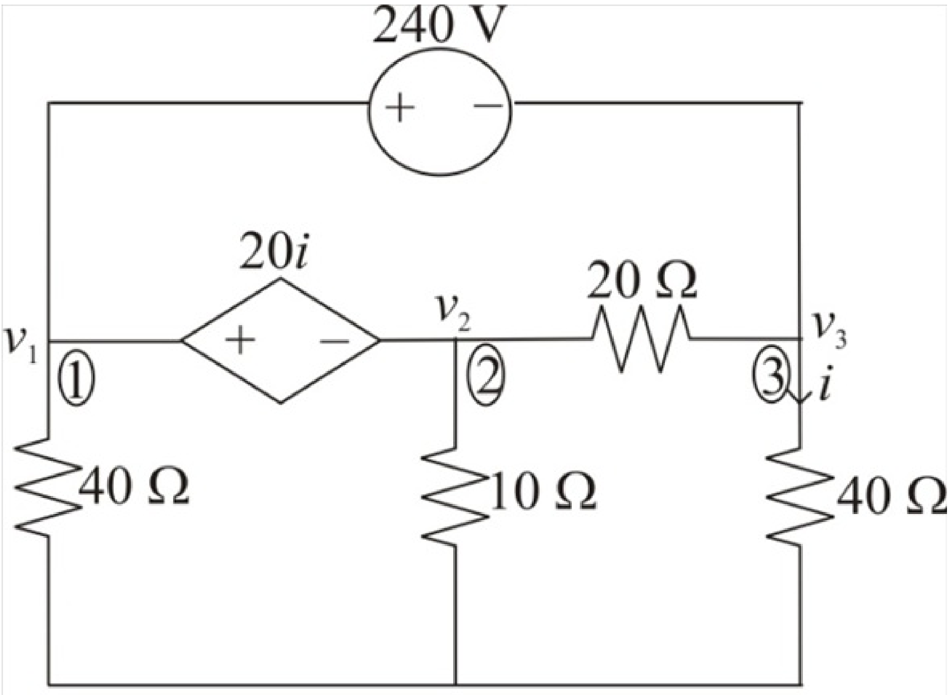


Figure 1

Step 2 of 10

Notice that the nodes 1, 2 and 3 represent the super node.

Consider the relation between the voltages v_1 and v_3.

$v_1 - v_3 = 240$

$v_3 = -240 + v_1$ …… (1)

Therefore, the expression for the voltage v_3 is $-240 + v_1$.

Step 3 of 10

Consider he expression for the current i.

$$i = \frac{v_3}{40}$$

Therefore, the expression for the current i is $\frac{v_3}{40}$.

Step 4 of 10

Consider the relation between the voltages v_1 and v_2.

$$v_1 - v_2 = 20i$$

Substitute $\frac{v_3}{40}$ for i.

$$\begin{aligned} v_1 - v_2 &= 20\left(\frac{v_3}{40}\right) \\ &= \frac{v_3}{2} \end{aligned}$$

Substitute $-240 + v_1$ for v_3.

$$v_1 - v_2 = \frac{(-240 + v_1)}{2}$$

$$v_1 - v_2 = -120 + \frac{v_1}{2}$$

$$v_2 = 120 - \frac{v_1}{2} + v_1$$

$$v_2 = \frac{v_1}{2} + 120 \quad \ldots\ldots \text{(2)}$$

Therefore, the expression for the voltage v_2 is $\frac{v_1}{2} + 120$.

Step 5 of 10

Apply super node analysis at nodes 1, 2 and 3.

$$\frac{v_1}{40} + \frac{v_2}{10} + \frac{v_2 - v_3}{20} + \frac{v_3 - v_2}{20} + \frac{v_3}{40} = 0$$
$$\frac{v_1 + 4v_2 + 2v_2 - 2v_3 + 2v_3 - 2v_2 + v_3}{40} = 0$$
$$v_1 + 4v_2 + 2v_2 - 2v_3 + 2v_3 - 2v_2 + v_3 = 0$$
$$v_1 + 4v_2 + v_3 = 0$$

Substitute $\frac{v_1}{2} + 120$ for v_2 and $-240 + v_1$ for v_3.

$$v_1 + 4\left(\frac{v_1}{2} + 120\right) + (-240 + v_1) = 0$$
$$v_1 + 2v_1 + 480 - 240 + v_1 = 0$$
$$4v_1 + 240 = 0$$
$$4v_1 = -240$$
$$\begin{aligned} v_1 &= -\frac{240}{4} \\ &= -60 \text{ V} \end{aligned}$$

Therefore, the value of the voltage v_1 is, $\boxed{-60 \text{ V}}$.

Step 6 of 10

Recall equation (1).

$$v_3 = -240 + v_1$$

Substitute -60 V for v_1.

$$\begin{aligned} v_3 &= -240 - 60 \\ &= -300\text{ V} \end{aligned}$$

Therefore, the value of the voltage v_3 is, $\boxed{-300\text{ V}}$.

Step 7 of 10

Recall equation (2).

$$v_2 = \frac{v_1}{2} + 120$$

Substitute -60 V for v_1.

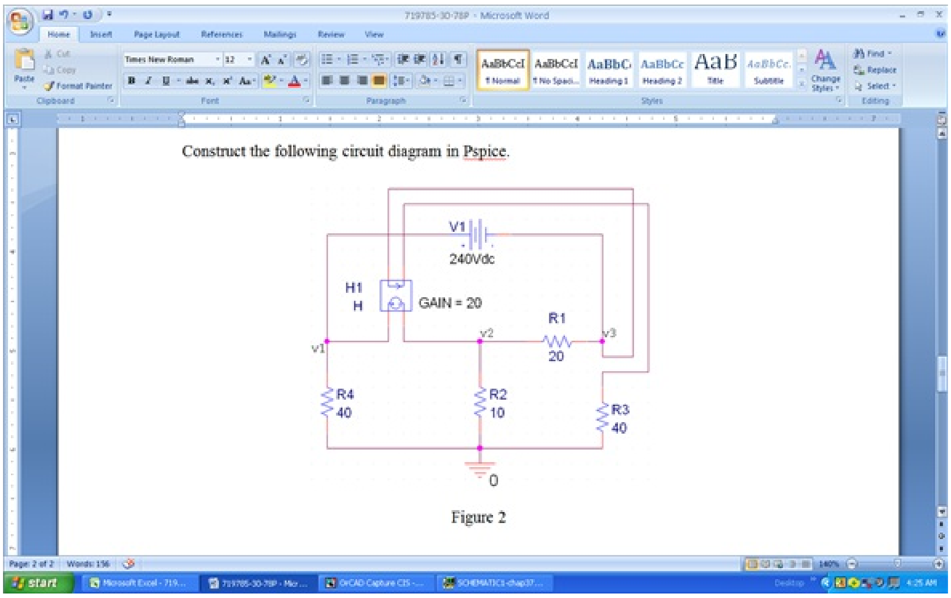
$$\begin{aligned} v_2 &= \frac{-60}{2} + 120 \\ &= -30 + 120 \\ &= 90\text{ V} \end{aligned}$$

Therefore, the value of the voltage v_2 is, $\boxed{90\text{ V}}$.

Step 8 of 10

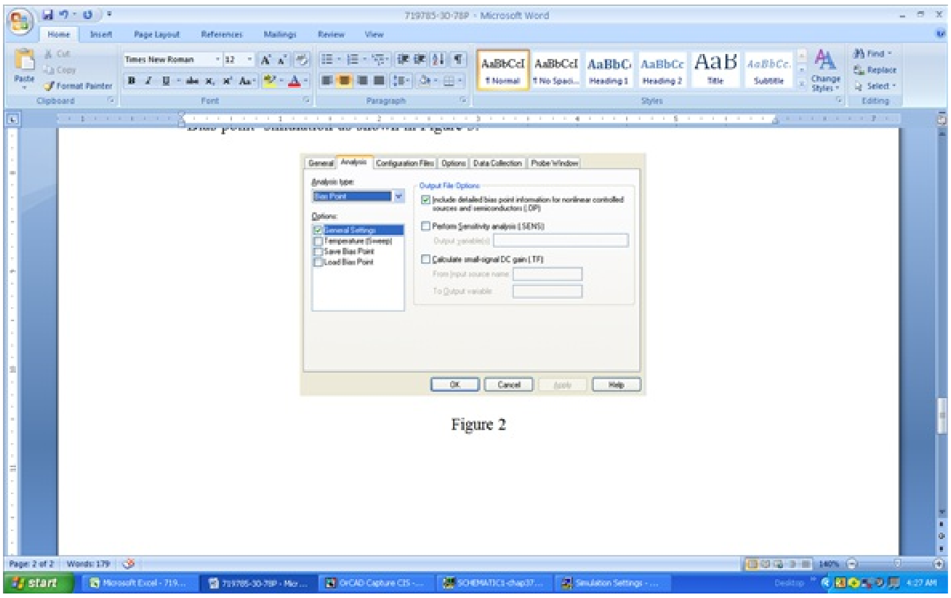
Pspice simulation:

Construct the following circuit diagram in Pspice.



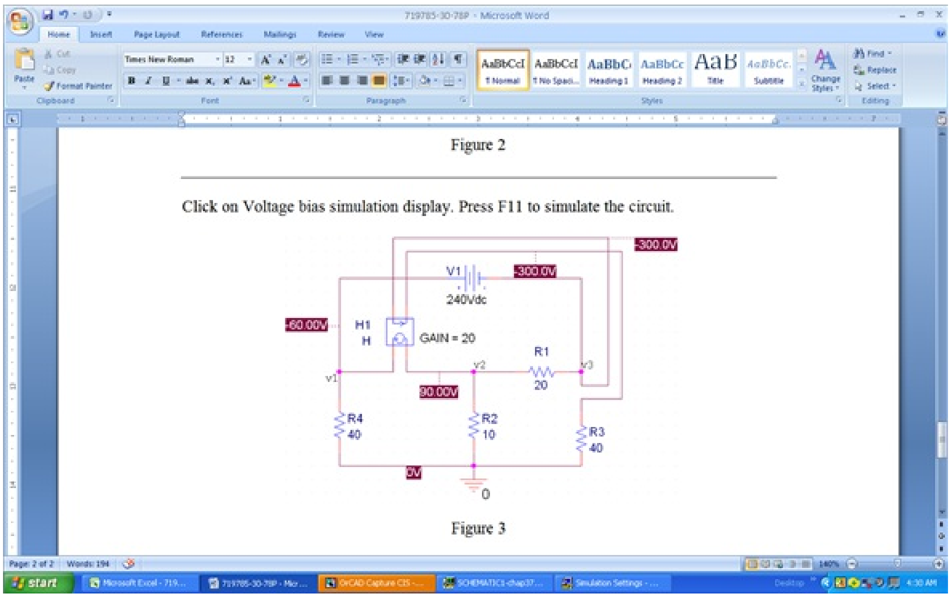
Step 9 of 10

Go to Pspice and create a new simulation profile. Edit the simulations profile and select 'Bias point' simulation as shown in Figure 3. Click on ok.



Step 10 of 10

Click on Voltage bias simulation display. Press F11 to simulate the circuit.



Therefore, the values of the voltages v_1, v_2 and v_3 are $\boxed{-60\text{ V}}$, $\boxed{90\text{ V}}$ and $\boxed{-300\text{ V}}$ respectively.